COMP6008 Reinforcement Learning
Group Project, Presentation, and Report Guidelines

July 29, 2026

1 Important

- This version of the assignment specification may be replaced, in whole or in part, by updates posted to Blackboard, either as new versions, as updates on the assignment page and/or as announcements posted to the Blackboard Announcements page.
  - The version date appears at the title of assignment specification (this document) and on the Blackboard Assessments/Project page.
  - Please make sure you follow the latest assignment specification as it may affect what you are expected to turn in.
  - No accommodation will be made for marks lost because you followed an outdated assignment specification.
- The project report will be checked through Turnitin.
  - The similarity with existing materials should be less than **15%** (excluding references and cover page), otherwise it will be treated as plagiarism and zero mark will be received for your project.
  - Please note NO similarity report will be available for the students after submission.
- **Timeline**:
  - Project Implementation and Presentation: 23:59, October 23, 2026 (Friday, Week 14).
  - Project Report (Turnitin submission): 23:59, October 30, 2026 (Friday, Week 15).

2 Objectives

In this project, students will work in groups to apply Reinforcement Learning (RL) technique(s) to solve specific problem(s), critically analyse the results, and present both the findings and the analysis. Additionally, students need to propose potential improvements or expansions to the existing work and provide justification for their proposals. The project aims to achieve the following learning outcomes:

- Gain a solid understanding of core concepts in reinforcement learning, such as Markov Decision Process, reward functions, value functions, policy, exploration-exploitation trade-offs.

- Learn how to model complex decision-making problems as Markov Decision Processes (MDPs), including defining the state space, action spaces, and designing reward function, and implement them in code.
- Develop hands-on skills in implementing (or adapting) and fine tuning reinforcement learning algorithms covered in the unit, and understand the mathematical foundations behind these algorithms.
- Acquire the experience of selecting appropriate reinforcement agents to solve complex decision making problems, and evaluating their performance through metrics such as average return(or reward), convergence speed, and sample efficiency, generalisation to unseen environments.
- Develop proficiency in identifying issues that arise during the implementation and training phases, and resolving these issues or proposing potential solutions with sound justification.
- Practice documenting project progress, including code documentation, experimental results, and insights gained, and effectively communicate findings through presentations or reports to both technical and non-technical audiences.
- Improve skills in group project planning, time estimation, task prioritisation, and resource allocation to ensure the timely completion of project milestones and deliverables.

3 Definitions

- Scientific justification: It means explaining why you are making a particular choice or claim, supported by credible scientific evidence. This should be based on peer-reviewed academic references such as journal articles, conference papers, or authoritative textbooks. Sources like blogs, Wikipedia, or unverified websites are not acceptable as evidence. For example, if you choose a particular algorithm, you should explain why it is appropriate and cite studies or authoritative sources that demonstrate its effectiveness.
- Scientific reasoning: It means showing the logic behind your arguments and linking the evidence to your conclusions. This requires you to explain how the referenced evidence supports your decisions, results, or interpretations. It also involves making assumptions explicit, considering alternatives, and ensuring your explanation is consistent with established scientific principles.

4 Summary of Project Requirements

1. Form a group of 3 students for your project study.
   - Only in exceptional cases, different group size may be permitted with prior approval from unit coordinator.
   - Sign up your group through:
     - Bentley campus: z
2. Project tasks
   - Select a real-world problem (which may be sourced from your employer), and get the approval from unit coordinator.
   - Reformulate the problem using RL framework.
   - Select and implement at least THREE suitable RL agents to solve it.
   - Analyse the performance.
3. Submit the well-commented project code along with documented instructions on how to run it. The submission worth 30 Marks and is due by 23:59, October 23, 2026 (Friday, Week 14).
4. Record a presentation of the project findings, with a maximum length of 20 minutes. The presentation worth 10 Marks and is due by 23:59, October 23, 2026 (Friday, Week 14)).
5. Write a group report that includes (due by 23:59, October 30, 2026 (Friday, Week 15))
   - Description and critical review of the project (worth 20 Marks).
   - Potential improvements AND expansion of existing work, with scientific justification (worth 15 Marks).

- Statements of individual members' contributions to the project are required. While no marks are allocated for this statement, failure to provide it will result in penalties for the group.

6. Finally, each group member submits a peer review report. This submission is optional if the group agrees that all group members contributed equally; otherwise, it is compulsory.

5 Project Selection

Students are expected to

- Select a real-world problem that is not trivial (that is, the solution is not obvious). The group is highly recommended to run the selected problem past the unit coordinator (UC) before going too much further into the project study to make sure that it isn't too simple or complicated. Otherwise, the group will lose marks and even fail the unit if the UC deems the selected topic as inappropriate.
- Formulate the selected problem as a RL environment. This formulation requires careful analysis and you can go through following questions before you start implementing the environment:
  - Does the environment emit observations?
    - If yes, what are the observation properties (continuous/discrete, dimensionality, range)?
    - If no, is it a multi-armed bandit problem?
  - Does the environment emit rewards?
    - If yes, are they suitable for RL (must be a number, not too sparse)?
    - If not, are you able to design suitable rewards that would allow RL agent to solve the environment?
  - Does the environment accept actions that affect its evolution?
    - If yes, what are the action properties (continuous/discrete, dimensionality, range)?
    - If no, it is not RL problem
  - How much time does it take for the environment to process one time step (receive action and emit observation and reward)? Remember that it may take RL agent millions of time steps to solve an environment.
  - Can you implement the environment in a reasonable time (avoid special algorithms/obscure data)?
- Implement the environment, i.e., the probability $p(s_{t+1}, r_{t+1}|s_t, a_t)$, in a Python notebook with at least following details
  - Observation space definition. Please note that the observation space defined here differs from the information observed by an agent at each time step. For instance, a robot's sensor may only output positional information at time step t as z_t. However, you can concatenate measurements from multiple time steps to define the observation space, such as $o_t = \{z_t, z_{t-1}, z_{t-2}\}$.
  - Reward signal emitted from environment.
  - Actions that an agent can take to impact the environment.
  - Transition model of the environment to the next state.

  It is recommended to use the gymnasium.Env class.
- Select THREE suitable RL agents for the chosen environment. The following guidelines apply:
  - You may reference existing RL libraries for ideas and understanding, such as:
    - Stable Baselines3 (link)
    - Ray RLlib (link)
    - Tianshou (link)

    Both Ray RLlib and Tianshou include offline RL algorithms, which are suitable for problems where only historical data is available (e.g., stock market data).
  - However, you must write your own implementation. You may study and draw inspiration from existing library code, and even use it as a starting point, but your final submission must be your own reimplementation that clearly reflects your understanding of the underlying theory. Simply calling library functions (e.g., `model = PPO(); model.learn()`) without reimplementation is not acceptable.

  - For each of the three agents, you must provide the pseudocode and explicitly link each step of the pseudocode to the corresponding part of your implementation.
  - Alternatively, you may adapt the practical implementations provided in this unit, provided they are suitable for solving the selected problem. The same reimplementation and pseudocode requirements above still apply.

- Perform multiple runs to solve the environment, including hyperparameter fine-tuning. It's possible that the RL agents may fail to solve the problem due to various factors beyond your control. However, it's crucial to note that your project can still achieve a high score if you provide a scientific justification for the failure and suggest potential improvements to the selected agent. Simply attributing failure to limited computational resources before the deadline is not sufficient.

6 Assignment

This assignment comprises two components: the project code submission in 6.1 and the presentation in 6.2.

6.1 Project Code (30 Marks)

The project code should be submitted before 23:59, October 23, 2026 (Friday, Week 14) and

- Prioritises readability and clarity.
  - Organise your code into a single, well-structured notebook.
    - Sections for each major component, such as environment setup, agent implementation, training loop, and result analysis.
    - Explanations for the logic flow between sections.
  - Organise the notebook with clear sections, the logic flow between sections should be smooth.
  - Use markdown cells to provide context, explanations, and instructions.
  - Code cells are accompanied by clear explanation and comments.
  - Present your results clearly using plots, graphs, or tables. Visualisations should be well-labelled and easy to interpret. Examples include learning curves (average reward over time), convergence plots, or tables comparing agent performance.
  - Demonstrate the process of hyperparameter fine-tuning and its impact on the agent's performance.
- Contains clear instructions for setting up the environment and reproducing results, if you are not using `Google Colab`.
  - include a "requirements.txt" file for your project (in the console type: pip freeze > requirements.txt).
  - include instructions to install dependencies and obtain additional data if they are required.
- Provides clear and well-structured mathematical description of the environment dynamics and pseudo-codes for the selected RL agents.
- Implements the chosen RL algorithms/methods comprehensively and correctly, ensuring that the code aligns with the provided pseudo-code and with the mathematical formulation of the environment process.
- Include the experimental code used to evaluate the effects of tuning critical hyperparameters (such as the learning rate) and to assess performance changes resulting from your modifications to the practical code or baseline algorithms.
  - Plots, graphs, or tables are preferred to demonstrate the results.

Marking rubric is provided in Table 1, and students are required to read it carefully.

Table 1: Project Code Marking Rubrics

Criteria	Excellent (≥85%)	Good (70-85%)	Satisfactory (50-69%)	Unsatisfactory (<50%)
Notebook organisation and structure (5%)	Notebook is well-organised with clear sections, logical flow between sections.	Notebook is organised with distinct sections but may lack some cohesion between them.	Sections are present but not well-structured. Flow between sections is unclear at times.	Notebook lacks organisation; sections are unclear or missing.
Clear Explanation of Code (10%)	Code cells are accompanied by clear explanations and comments, also markdown cells are provided to aid understanding of the purpose and functionality of the code.	Code explanations are present but may lack clarity or detail in some parts.	Code explanations are brief or somewhat unclear. Some code cells lack context.	Limited code explanations; code cells are often lacking context.
Implementation of Reinforcement Learning (40%)	Comprehensive and correct implementation of chosen RL algorithms/methods and chosen environment modelling, and match between pseudo code and implemented algorithm (or environment model)	Minor errors may be present, but they don't significantly affect results.	Basic implementation with notable errors or omissions.	Implementation is incomplete, leading to errors in results.
Experimentation and Result Presentation (35%)	Thorough experimentation with comprehensive results. Clear presentation using plots, graphs, or tables.	Reasonable experimentation with clear presentation of results. Some minor gaps in the presentation.	Limited experimentation with unclear presentation of results.	Minimal experimentation and unclear presentation.
Reproducibility and Environment Setup (10%)	Clear instructions for setting up the environment and reproducing results. Dependencies are clearly listed.	Instructions for environment setup and result reproduction are present, but some details may be lacking. Dependencies are mostly listed.	Setup instructions are somewhat unclear. Dependencies may be missing or incomplete.	Setup instructions are unclear or inaccurate. Dependencies are not properly listed. The student will receive **ZERO** mark for project, and therefore **fails** the unit.
Note 1: If `Google Colab` is used, the criterion *Reproducibility and Environment Setup* will not be assessed. Instead, marks for the remaining criteria will be scaled up to a total of 30. However, a zero mark penalty will still apply if the results presented in the presentation and report are not included in the notebook.				

6.2 Presentation (10 Marks)

Each group is required to submit a recorded presentation before 23:59, October 23, 2026 (Friday, Week 14). It is recommended to use PowerPoint and instructions are available here. The presentation should strictly adhere to a maximum duration of 15-20 minutes per group (penalties will be applied for exceeding this time limit) and include ONLY:

- A brief description of the problem to be solved, and explain
  - why the problem is important?
  - why it is challenging/difficult?
  - why RL is suitable to solve the problem?
- Give a detailed explanation of your design and implementation of the environment that represents the problem, it should include

  - environment state definition.
  - reward design and why it match the final goal to be achieved by RL agents.
  - action space definition.
  - mathematical description of the environment dynamic model $p(s'|s,a)$ OR data source used to simulate the model $p(s'|s,a)$.

  Please note that the presentation should also include the linkage between code snippets and the theories that guide your implementation.

- A brief description of the RL agents that you plan to deploy, and explain why the agents are selected, i.e., why the agent can solve the problem.
- Present the training results using visual aids such as learning curves showing the average cumulative reward per episode over time, convergence plots, and tables comparing agent performance, etc.
- Marking rubric for this option are available in Table 2.

Table 2: Presentation Marking Rubrics

Criteria	Excellent (≥85%)	Good (70-85%)	Satisfactory (50-69%)	Unsatisfactory (<50%)
Problem Description (15%)	Thoroughly and concisely explains the problem's significance and challenges, demonstrating deep understanding and engaging the audience.	Clearly explains the problem's importance and challenges, showing a good understanding.	Describes the problem's importance and challenges in a basic manner, with noticeable gaps.	Provides a vague or unclear description of the problem's importance and challenges.
Environment design and implementation (35%)	Presents a comprehensive, well-designed environment reflecting a deep understanding of the problem.	Presents a clear environment design and implementation, demonstrating an effective grasp of the problem.	Presents the environment design and implementation but with minor omissions or less clarity.	Presents an environment design that is incomplete or significantly lacks clarity in implementation.
RL agent descriptions (20%)	Provides detailed and insightful explanations of the RL agents, showcasing a strong grasp of their functioning and relevance to the problem.	Offers clear explanations of the RL agents with a good understanding of their role in solving the problem.	Describes the RL agents with noticeable gaps in understanding or relevance to the problem.	Provides vague or inaccurate descriptions of the RL agents, indicating a significant lack of understanding.
Results presentation (10%)	Training results are presented with clear, well-labeled learning curves, convergence plots, and/or comparative tables. Visuals are accurately interpreted, and key insights are discussed (e.g., convergence speed, stability, agent performance). Results support conclusions in the report.	Includes useful plots or tables showing training progress and agent performance. Visuals are mostly clear and labelled, with basic interpretation. Some insights are mentioned, though limited in depth.	Includes training results with basic learning curves or tables, but may lack clarity, labels, or sufficient explanation. Interpretation is minimal or superficial.	Includes incomplete or unclear plots/tables with little to no interpretation. Visuals may be misleading or not relevant to performance evaluation.
Clarity(20%)	Speaks clearly and articulately throughout the presentation, ensuring all content is easily understood; complete within the time limits;	Generally speaks clearly, with only minor instances of unclear speech.	Speech is mostly clear, but with some instances of unclear delivery.	Consistently unclear speech, significantly impeding understanding of the presentation.

7 Project Report (35 Marks)

The project report should be split into three parts: (1) a description and critical review of the project, (2) proposals for potential improvements and expansion of the existing work, and (3) a statement of individual contributions, and it should be submitted before 23:59, October 30, 2026 (Friday, Week 15).

7.1 Part 1: Description and Critical Review (20 Marks)

This part should be based what you have achieved throughout your project, and you need to

- Provide a background introduction to the selected problem, explain why it is important for the project to be solved, and what are the challenges to solve the problem using RL techniques. The importance of the project can be highlighted by outlining the benefits to society, or the potential contribution to expanding knowledge in the field of RL.

- Give a detailed description of the Markov decision model for the selected problem, include
  - The state and action spaces design with scientific justification.
  - Reward signal design and explain why shaped reward signal will help to achieve the final goal of the selected problem and speed up the training of RL agents.
  - State transition model from current state to next state. It can be mathematical model from physics law, simulation software, or data driven.
- Justify of selected agents, and describe of the process of tuning hyper-parameters.
- Analyse the training results, and validate the agent in the testing environment or testing data.
- Critically review the projects work. For examples, has the agent solved the problem? If yes, which part of your work contribute to the success. If not, what are the potential causes to the failure? What are the implication of achieved results?
- The marking rubric is available in Table 3.

Table 3: Marking Rubrics of Project Report - Part 1

Criteria	Excellent (≥85%)	Good (70-85%)	Satisfactory (50-69%)	Unsatisfactory (<50%)
Problem Description (10%)	Comprehensive description of the problem addressed, including its significance and relevance. Clearly articulates the challenges to be overcome.	Problem description is presented but lacks depth or clarity. Importance may not be fully conveyed.	Problem description is somewhat unclear. Importance and challenges are vaguely stated.	Problem description is missing or unclear. No context, importance, or challenges presented.
Environment Model (25%)	Detailed explanation of the environment model that relevant attributes, states, actions and rewards, are correctly defined with scientific justification.	Clear environment description with relevant attributes, states, actions, and rewards. Some minor details might be missing.	Environment description is present but lacks clarity or completeness for some parts of the problem definition.	Environment description is vague or lacks important details.
Agent Training (10%)	Comprehensive discussion of the process of choosing agents and tuning agent parameters. Clear reasoning behind parameter choices and their impact on performance.	Clear explanation of choosing agents and parameter fine-tuning process, though some aspects could be more detailed. Some reasoning behind choices is provided.	Processes of agent selection and parameter fine-tuning are outlined, but lacks depth or clarity. Limited reasoning for parameter choices.	Parameter fine-tuning process and agent selection are somewhat unclear. Little reasoning for choices provided.
Results and Outcome (25%)	Thorough presentation of results, including quantitative and qualitative outcomes. Clear insights drawn from the results and their relevance to the problem and chosen agents.	Clear presentation of results with insights drawn, though some analysis might be lacking depth. Relevance to the problem and chosen agents is evident.	Presentation of results is present, but some details may be unclear. Insights and analysis may lack depth.	Results are presented, but their relevance to the problem and chosen agents is not fully explained. Insights may be superficial.
Critical Review of Work (20%)	Thorough and thoughtful critical review of the project's work, including the match between expected and achieved results. Potential implications are deeply considered and discussed.	Comprehensive review of work, discussing alignment between expectations and outcomes, but depth may vary.	Basic review of work with some discussion on results and possible implications. Analysis lacking depth.	Limited critical review; discussion on results and implications is superficial.
Presentation and Clarity (10%)	Well-organized and visually appealing presentation of the problem, environment, agent(s), training, results, and review. Clear explanations with minimal errors. Use the IEEE format without errors.	Neat presentation with organized content. Explanations are generally clear, though some sections might lack clarity. Minor errors in writing. Minimal inconsistency in IEEE format	Presentation is somewhat disorganised, affecting overall clarity. Explanations may be unclear in parts. Noticeable errors in writing. Major inconsistency with IEEE format.	Presentation is messy or inconsistent. Explanations lack clarity and are hard to follow. Significant errors in writing. Non IEEE format is used.

7.2 Part 2: Potential Improvements and Expansion (15 Marks)

In this part, students need to conduct research on existing literature to identify future directions and potential improvements for their existing project work. You will need to

- List the limitations and weaknesses of existing work in your project, and identify a set of potential improvements (at least as many as the group size of improvements) that could overcome these limitations.
  - the discussion of limitations and weakness should be evidenced by existing project work.
  - the potential improvements should be well-reasoned with the support of existing academic literature.
  - the tuning or modification of hyperparameters cannot be considered an improvement.
- Identify directions for future work (at least as many as the group size). Please note that these directions should NOT directly relate to overcoming the limitations and weaknesses identified in your project's results above. They should instead aim to extend the project's scope or enhance performance by introducing novel approaches not previously used in your work. Furthermore, these proposed future works should be supported by existing academic literature, and justified with scientific reasoning.
- Only peer-reviewed journal and conference articles, dissertations, and textbooks can be counted as academic literature. A minimum of SIX pieces of literature should be included for this part.
- The marking rubrics are available in Table 4.

This part will require substantial effort, and achieving high marks will depend on conducting extensive reading and research. It is recommended that at least 40% of your report be dedicated to this part.

Table 4: Marking Rubrics of Project Report - Part 2

Criteria	Excellent (≥85%)	Good (70-85%)	Satisfactory (50-69%)	Unsatisfactory (<50%)
Potential Improvements (45%)	Provides a comprehensive and well-reasoned list of potential improvements that could enhance the project. Improvements are directly linked to identified weaknesses and limitations.	Lists potential improvements but without strong links to identified weaknesses. Improvements may lack depth or clarity.	Suggests basic potential improvements, but the connections to identified limitations are unclear or weak.	Lists few or irrelevant potential improvements. Limited connection to identified weaknesses.
Future Work Directions (45%)	Presents insightful and well-explained directions for future work that could extend the project's scope or boost the performance. Clear relevance to the project's goals.	Offers meaningful suggestions for future work that could expand the project's scope, a brief justification is provided but some relevance to the project's goals might be missing.	Provides some directions for future work, but they may be somewhat generic or not closely tied to the project's goals and justification is minimal.	Suggests future work directions, but their relevance or justification is not reasonable.
Presentation and Clarity (10%)	Writing is clear, concise, and easily understood. Sentences and paragraphs are well-structured. Cohesion between ideas and paragraphs is evident. Use IEEE format without errors.	Writing is generally clear and understandable, but some sentences or paragraphs may be verbose or lack clarity. Cohesion could be improved. Minimal inconsistency in IEEE format.	Writing is somewhat unclear or verbose in places. Paragraphs lack clear connection. Cohesion may be weak. Major inconsistency with IEEE format.	Writing is often unclear, making it difficult to understand the ideas presented. Lack of cohesion between paragraphs. Non IEEE format is used.

7.3 Part 3: Statement of Contributions

The purpose of this section is to ensure fair and transparent marking for both the **Project Code** and the **Project Report**, based on each group member's individual contribution.

- Each section of the group project code and project report must be individually attributed to a specific group member. The recommended approaches are:

  - Report: Include a footnote or parenthetical note at the beginning of each section, e.g., "(Written by Student A)".
  - Code: Use a Markdown cell at the beginning of each section in the Jupyter notebook to identify the contributor(s), e.g., `**Author: Student A**`.

- As a group, agree on each member's contribution, quantified as a percentage. For example: Student A (40%), Student B (35%), and Student C (25%).
  - In most cases, the following bounded modifiers will be applied based on each member's percentage contribution. The example below assumes a group of 3 students:
    * $< 10\%$ contribution $\rightarrow$ up to -100% deduction.
    * $\geq 10\%$ and $< 15\%$ contribution $\rightarrow$ up to -70% deduction.
    * $\geq 15\%$ and $< 20\%$ contribution $\rightarrow$ up to -50% deduction.
    * $\geq 20\%$ and $< 25\%$ contribution $\rightarrow$ up to -40% deduction.
    * $\geq 25\%$ and $< 33.3\%$ contribution $\rightarrow$ up to -25% deduction.
    * $\geq 33.3\%$ contribution $\rightarrow$ up to a bonus of $(3 \times \text{individual contribution} - 100\%) \times$ total marks in percentage.
  - The individual mark is calculated as:

$$\text{Individual Mark} = \text{Group Mark} \times (1 + \text{Modifier})$$

    Examples:
    * A student with a 20% contribution and a group mark of 70 receives a modifier of -40%, resulting in an individual mark of $70 \times (1 - 0.40) = 42$.
    * A student with a 40% contribution and a group mark of 70 receives a bonus modifier of up to $20\% \times 70/100 = 14\%$, resulting in a maximum individual mark of $70 \times (1 + 0.14) = 80$.
  - If all group members unanimously agree that contributions were equally distributed, a simple statement suffices: "All group members contributed equally to this project." In this case, all members will receive the same mark.
  - If the group cannot reach an agreement, individual members may submit the requested confidential Peer Review Report, uploaded in
  - The unit coordinator reserves the right to review all peer evaluations. Evaluations deemed overly harsh or excessively generous may be adjusted at the unit coordinator's discretion.

- Each student's final mark will be determined based on the marker's assessment of the report's attribution statements and all group members' Peer Review Reports.

7.4 Report Format

The report must be no more than **eight** A4 pages in IEEE two-column format. This page limit excludes code listings, references, and mathematical derivations, which should be placed in an appendix on separate pages.

- Templates in both MS Word and LaTeX format are available at:
  `https://www.ieee.org/conferences/publishing/templates.html`.
  Do not alter the template's margins, fonts, or spacing to fit additional content. Your effort is better spent making the report more concise and clearly written.

- The required referencing style is IEEE style. Detailed guidance on this style is available at:
  `https://ieeeauthorcenter.ieee.org/wp-content/uploads/IEEE-Reference-Guide.pdf`.